Assignment 1

1

a

1

$$\Theta(n^2)$$

2

$$\Theta(n^2)$$

3

$$\Theta(n \log(n))$$

4

$$\Theta(n)$$

b

a: loop conditional
b: appending
c: linear search
d: initialization

$$f_1(n) = d + \sum_{i=1}^{n}(a + c(i-1) + b)$$

a: loop conditional
b: sorted insertion
c: binary search
d: initialization

$$f_2(n) = d + \sum_{i=1}^{n}(a + c\log(i-1) + b(i-1))$$

a: loop conditional
b: insertion

c: contains
d: initialization

$$f_3(n) = d + \sum_{i=1}^{n}(a + c\log(i-1) + b\log(i-1))$$

a: loop conditional
b: insertion
c: contains
d: initialization

$$f_4(n) = d + \sum_{i=1}^{n}(a + c + b)$$

C

$$f_1(n) = d + \sum_{i=1}^{n}(a + c(i-1) + b)$$

$$f_1(n) = d + \sum_{i=1}^{n} a + \sum_{i=1}^{n} b + c\left(\sum_{i=1}^{n} i - \sum_{i=1}^{n} 1\right)$$

$$f_1(n) = d + na + nb + c\left(\frac{n(n+1)}{2} - n\right)$$

$$\boxed{f_1(n) = d + n\left(a + b + c\left(\frac{n+1}{2} - 1\right)\right)}$$

$$f_2(n) = d + \sum_{i=1}^{n}(a + c\log(i-1) + b(i-1))$$

$$f_2(n) = d + \sum_{i=1}^{n} a + b\left(\sum_{i=1}^{n} i - \sum_{i=1}^{n} 1\right) + c\sum_{i=1}^{n}\log(i-1)$$

$$\sum_{i=1}^{n}\log(i-1) \leq \sum_{i=1}^{n}\log(n-1)$$

$$\sum_{i=1}^{n}\log(i-1) \leq (n-1)\log(n-1) = O(n\log n)$$

$$\sum_{i=1}^{n}\log(i-1) \geq \sum_{i=\lceil\frac{n}{2}\rceil}^{n}\log(i-1)$$

$$\sum_{i=1}^{n} \log(i-1) \leq \frac{n}{2} \log\left(\frac{n}{2}\right) = \Omega(n \log n)$$

$$f_2(n) = d + na + b\left(\frac{n(n+1)}{2} - n\right) + c\Theta(n \log n)$$

$$\boxed{f_2(n) = d + n\left(a + b\left(\frac{n+1}{2} - 1\right)\right) + c\Theta(n \log n)}$$

$$f_3(n) = d + \sum_{i=1}^{n} (a + c\log(i-1) + b\log(i-1))$$

$$f_3(n) = d + \sum_{i=1}^{n} a + b\sum_{i=1}^{n} \log(i-1) + c\sum_{i=1}^{n} \log(i-1)$$

$$\boxed{f_3(n) = d + na + b\Theta(n \log n) + c\Theta(n \log n)}$$

$$f_4(n) = d + \sum_{i=1}^{n} (a + c + b)$$

$$f_4(n) = d + \sum_{i=1}^{n} a + \sum_{i=1}^{n} b + \sum_{i=1}^{n} c$$

$$f_4(n) = d + na + nb + nc$$

$$\boxed{f_4(n) = d + n(a + b + c)}$$

2

$4 < 3 < 1 < 2$
4 is the fastest algorithm

3

$$f_1(500000) = 1 + 500000\left(1 + 1 + 1\left(\frac{500000+1}{2} - 1\right)\right)$$

$$\boxed{f_1(500000) = 1.2500075 \cdot 10^{11}}$$

$$f_2(500000) = 1 + 500000\left(1 + 1\left(\frac{500000 + 1}{2} - 1\right)\right) + 1\Theta(500000 \log 500000)$$

$$\boxed{f_2(500000) = 1.2500309949 \cdot 10^{11}}$$

$$f_3(500000) = 1 + 500000(1) + 1\Theta(500000 \log 500000) + 1\Theta(500000 \log 500000)$$

$$\boxed{f_3(500000) \approx 6198971}$$

$$f_4(500000) = 1 + 500000(1 + 1 + 1)$$

$$\boxed{f_4(500000) = 1500001}$$